Pollock (*Pollachius virens*)

Summary Scores

- **Sensitivity:** High
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.12	2.73	
Complexity in Reproductive Strategy	3.44	2.5	
Dispersal of Early Life Stages	1.08	3	
Egg and Larval Survival and Settlement Requirements	2.96	1.7	
Historical Variability in Temperature	2.48	3	
Juvenile Habitat Requirements	2.44	3	
Other Stressors	1.68	2.2	
Population Growth Rate	3.56	2.2	
Prey Specificity	2.08	2.75	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.68	1.4	
Spawning Cycle	2.36	2.6	
Stock Size/Status	1.12	3	
Total Sensitivity | Certainty	**High (3) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.31	1.08	
Bottom Oxygen	2.92	0.94	
Bottom Salinity	2.32	1.06	
Bottom Aragonite Solubility	1.21	0.55	
Small zooplankton integrated biomass	1.26	0.26	
Medium zooplankton integrated biomass	1.42	0.27	
Large zooplankton integrated biomass	1.4	0.39	
Total Exposure	**Moderate (1.93)**	**0.92**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (5.8)**	**2.76**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Pollock. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Moderate	High	Increase
Exposure	High	Moderate	Decrease
Vulnerability	Moderate	Moderate	None
Directional Effect	Negative	Negative	None

Stocks

Pollock is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

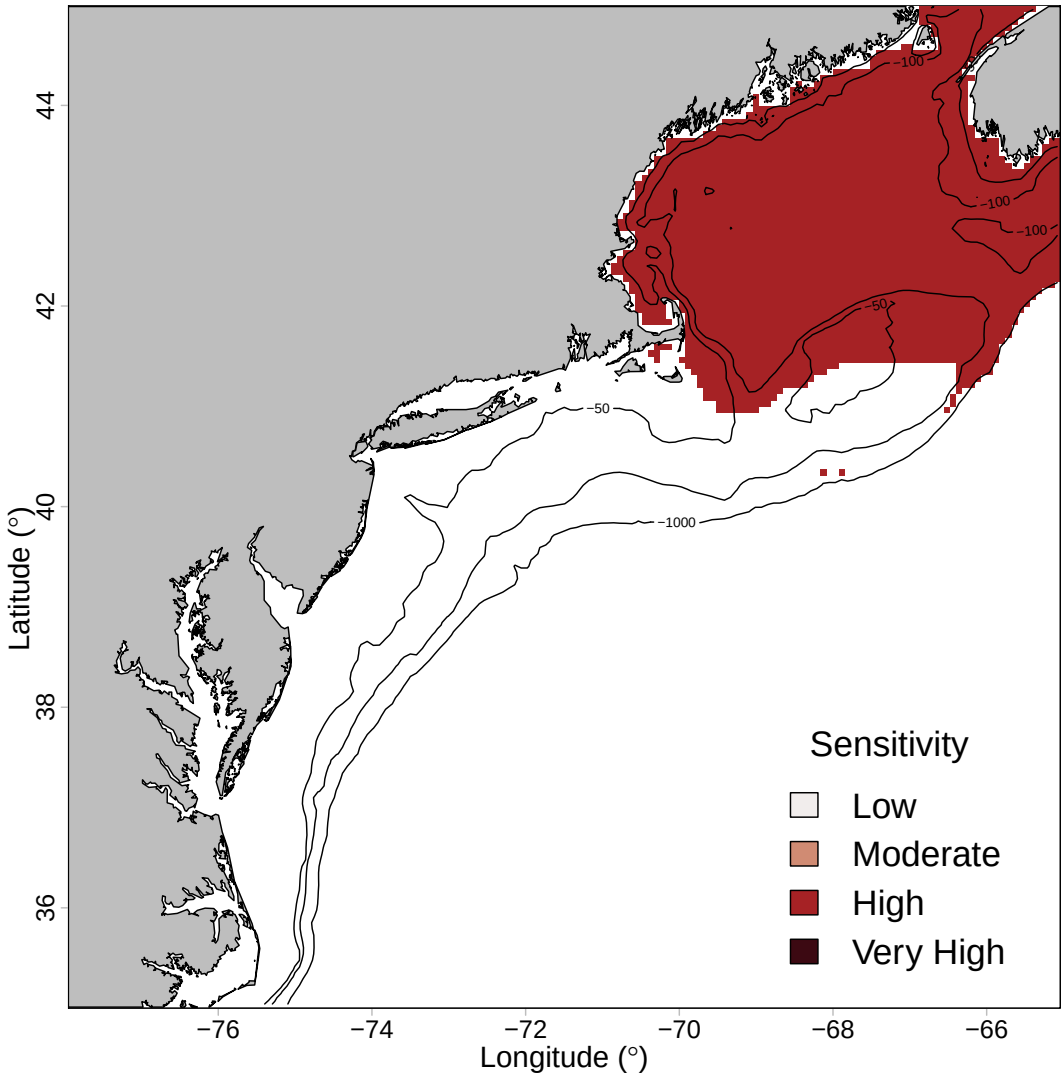


Figure 2: Pollock Sensitivity on the Northeast Shelf

Pollock received a **High** sensitivity score, with 100% certainty according to bootstrap analysis. Two attributes had scores greater than 3: Population Growth Rate (3.6) and Complexity in Reproductive Strategy (3.4) (Figure 1). Pollock are long-lived and relatively slow to mature, reaching maturity at 4-7 years of age and living for up to 24 years (Northeast Fisheries Science Center (NEFSC) 2013). Spawning occurs at a narrow range of temperatures (Cargnelli et al. 1999) (Collette and Klein-MacPhee 2002). Sound production, and conversely the ability to hear sounds, may play a role in reproduction (Looby et al. 2022) (Hawkins and Picciulin 2019).

In the 2016 Climate Vulnerability Assessment, Pollock received a Moderate sensitivity score. This was driven by Population Growth Rate, similar to our study, and Early Life History Requirements, because juveniles rely on inshore habitats (Hare et al. 2016). Our corresponding attribute, Egg and Larval

Survival and Settlement Requirements, received a score of 3 in this assessment, up slightly from the previous assessment (Figure 1).

Exposure:

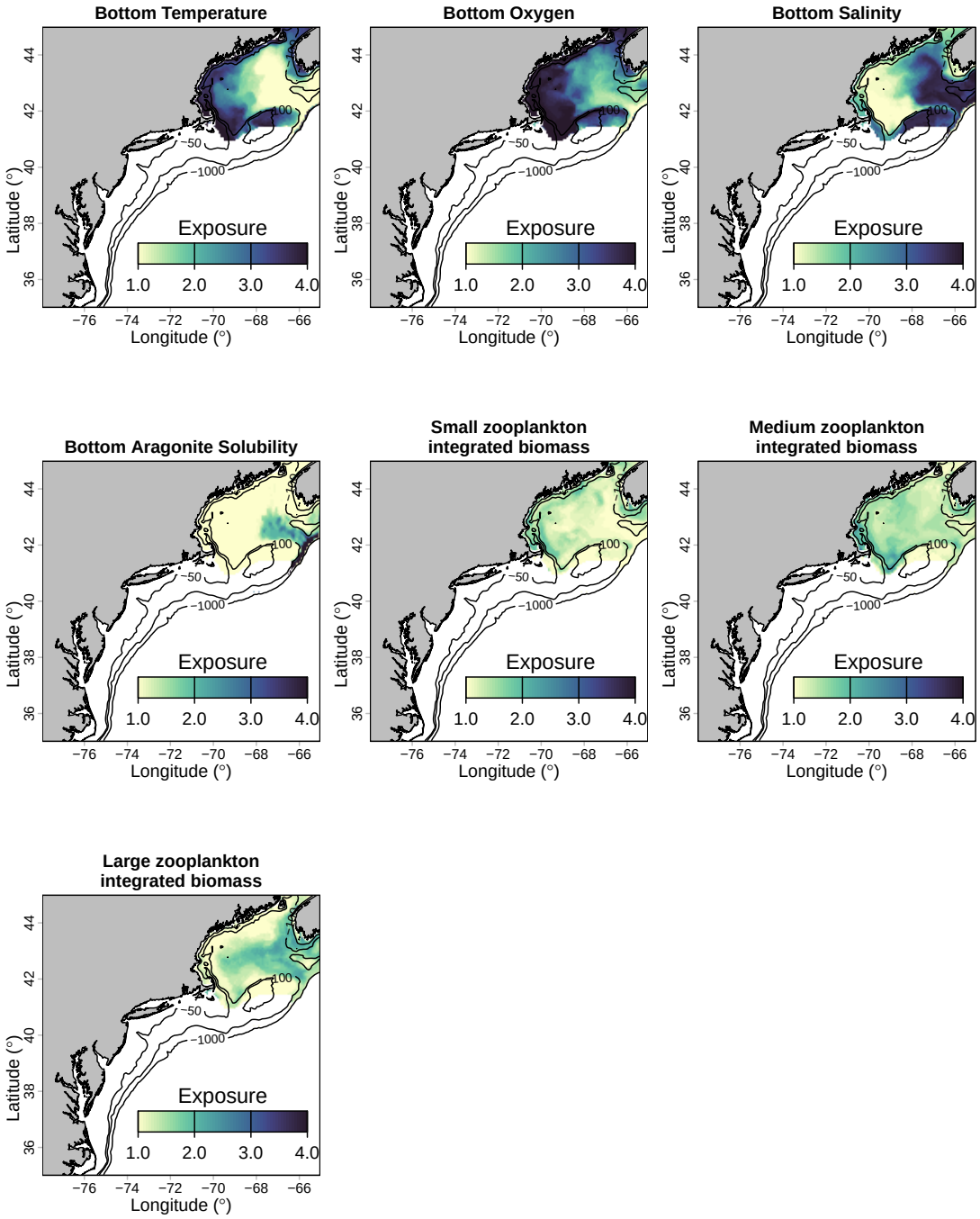


Figure 3: Maps of each Exposure Factor for Pollock. Exposure was calculated between 2014-2023 and 2025-2035

Mean exposure for Pollock was **Moderate** (1.93 +/- 0.92) (Figure 1). Scores were Very High over Georges Bank and High along the coast of the Gulf of Maine. Scores were Low away from the coast (Figure 4). These patterns were driven by high bottom temperature and oxygen exposures (Figure 3). Pollock received a High score in the previous Assessment, driven by sea surface temperature and ocean acidification exposures (Hare et al. 2016).

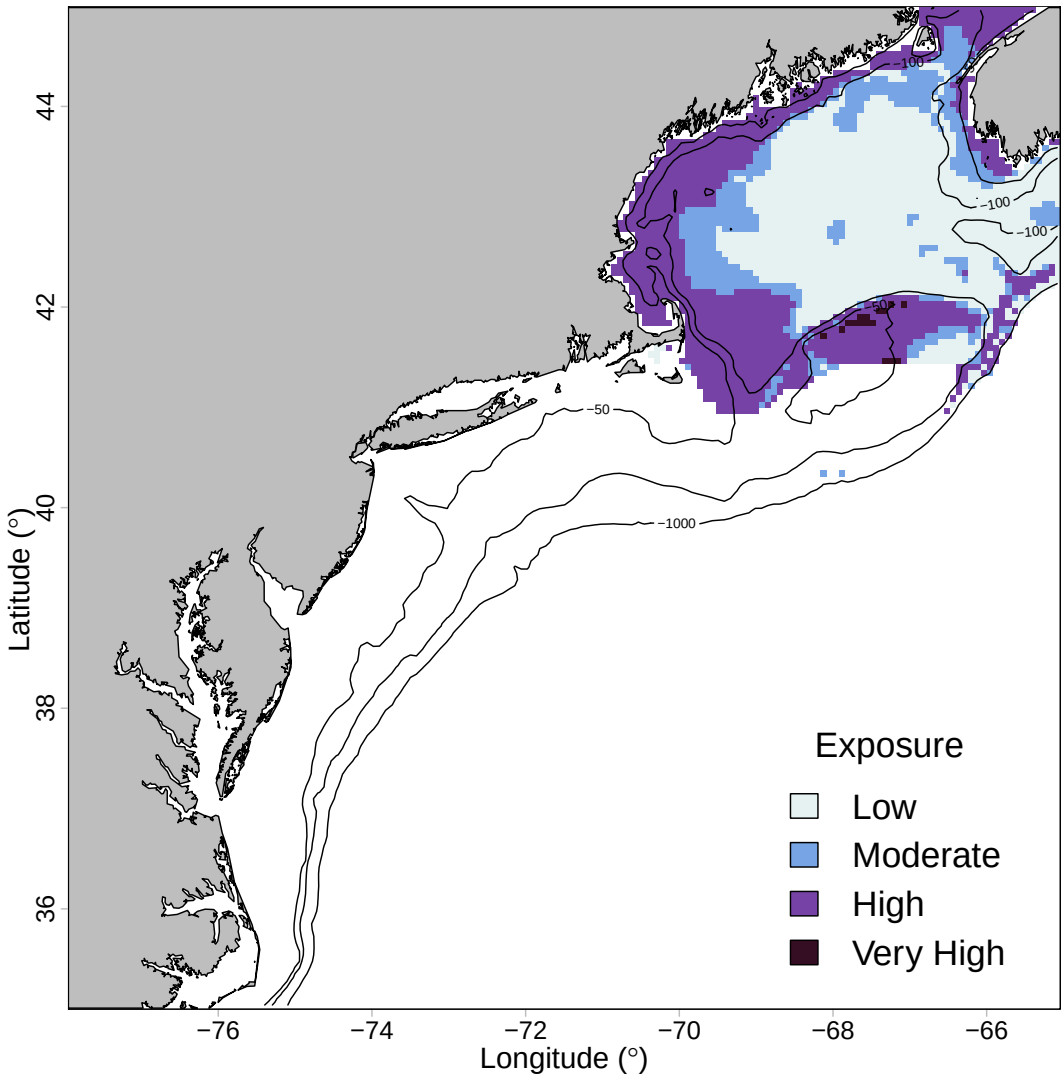


Figure 4: Total Exposure on the Northeast Shelf for Pollock. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

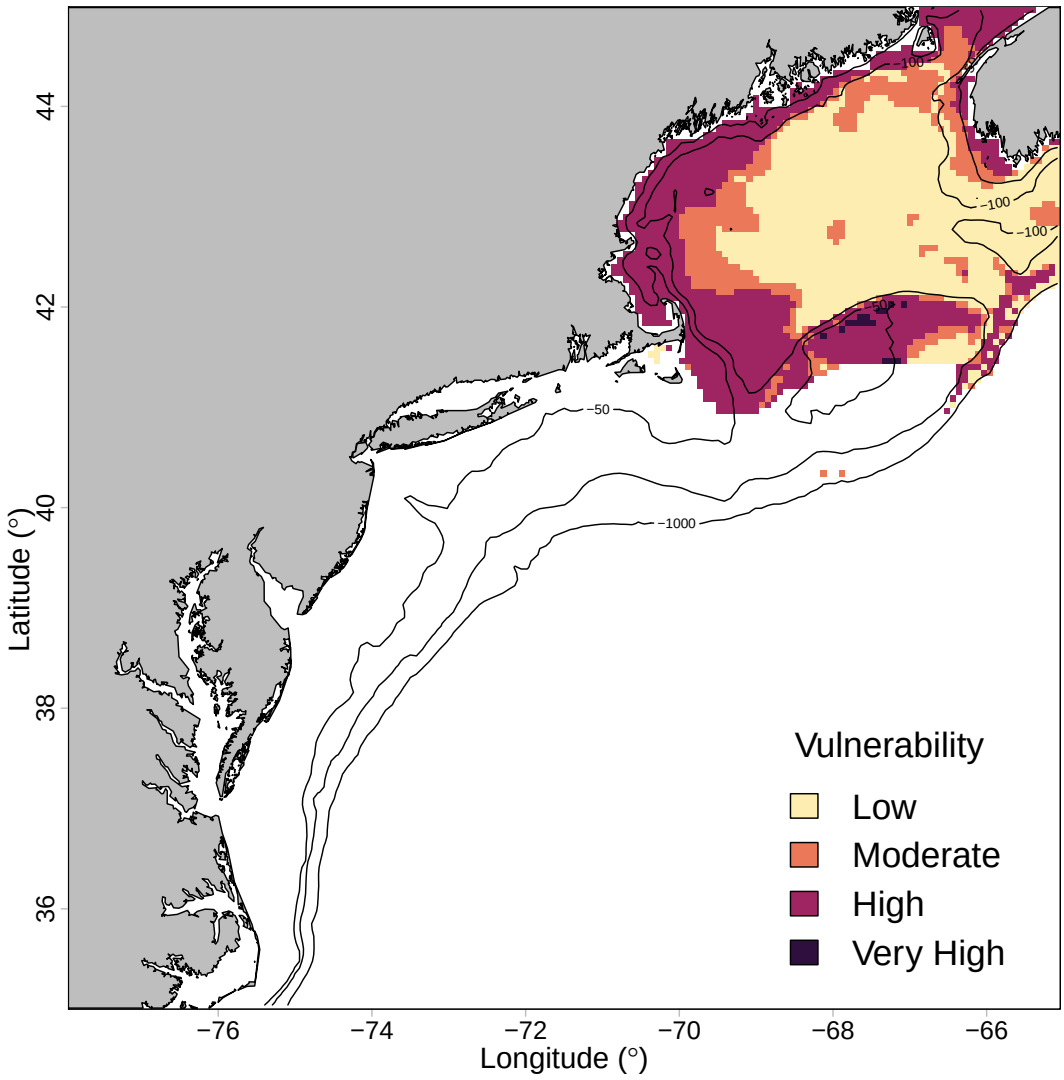


Figure 5: Vulnerability on the Northeast Shelf for Pollock. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Pollock was **Moderate** (5.8 +/- 2.76) (Figure 1). Vulnerability was highest on Georges Bank (Very High), followed by the coast of the Gulf of Maine (High). Vulnerability decreased away from the coast (Figure 5).

In the previous Assessment, Pollock also received a Moderate Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

85% of data quality scores were greater than or equal to 2, indicating strong data quality. Egg and Larval Survival and Settlement Requirements and Relationship to Calcified Organisms were considered data gaps

(mean attribute data quality score < 2) (Figure 1).

Model Confidence Score:

The pollock ensemble model received a mean model confidence score of 2.4 (+/-0.5), indicating moderate confidence in the model. The model captured known pollock distributions well across time and space. The ensemble model AUC score was 0.90, indicating excellent model performance.

Additional Metrics

Distribution Shifts:

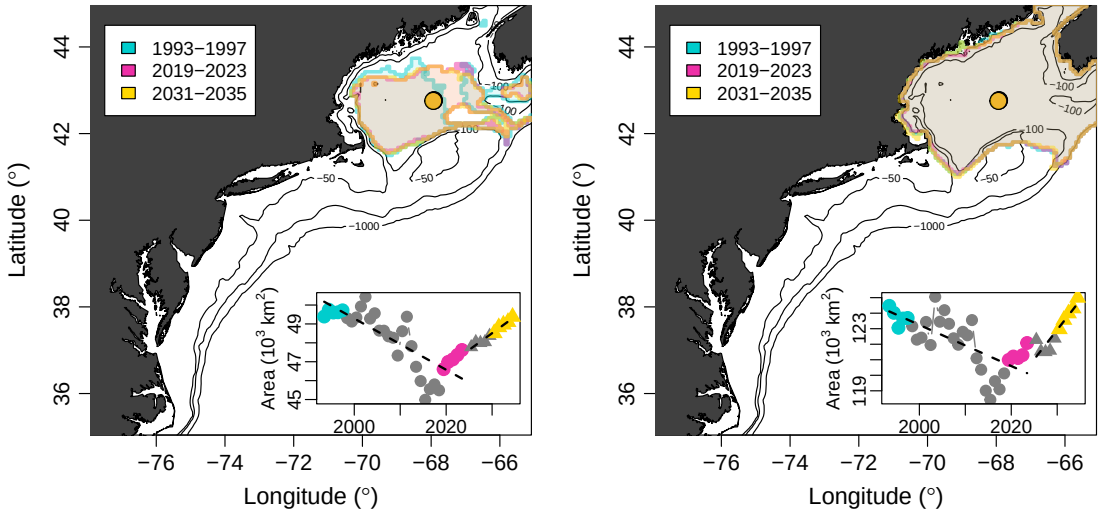


Figure 6: Distribution shifts on the Northeast Shelf for Pollock. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Pollock experienced significant declines in habitat area between 1993 and 2023, despite recent rebounds. Pollock core habitats moved deeper into the Gulf of Maine (Figure 6). The overall shift was minimal; the weighted centroid shifted less than 2 km to the south.

Under near future ocean conditions, pollock habitats are predicted to expand significantly, reaching near 1993 levels. This expansion is predicted to be uniform across the Gulf of Maine and the weighted centroid is predicted to remain stable (Figure 6).

In the previous assessment, pollock received a High Distributional Vulnerability Rank due to their high mobility and high larval dispersal (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	49,822	123,835	2.74	4.65
2019-2023	1.7 km S	47,165 (-2,656)	121,391 (-2,445)	2.68 (-0.06)	4.49 (-0.16)
2031-2035	1.7 km WSW	48,966 (+1,801)	123,988 (+2,595)	2.68 (0)	4.57 (+0.08)

Directional Effect:

Experts predicted that Pollock would be **negatively** impacted by climate change. Certainty was 70%. This is similar, albeit with lower certainty, than the results from the 2016 Assessment (Hare et al. 2016). Pollock spawn at a narrow range of temperatures and prefer temperatures between 3 and 11 C (Cargnelli et al. 1999) (Collette and Klein-MacPhee 2002) (Scott 1982). Shifts to deeper waters have already been observed (Nye et al. 2009) (Murawski 1993).

Climate Effects on Abundance and Distribution

Because of the narrow range for when spawning occurs (4.5-6°C) (Cargnelli et al. 1999) (Collette and Klein-MacPhee 2002), increase in temperatures and climate change could have implications on the stock's success. The center of biomass for Northwest Atlantic pollock has not shifted and distribution has not moved northward, although they have been recently found at greater depths (Nye et al. 2009). Murawski (1993) noted a range contraction, increased depth of occurrence, and increased temperature of occurrence in pollock and other deep-water fish in the Gulf of Maine.

Life History Synopsis

Atlantic pollock are found in the Northwest Atlantic from the Grand Banks and Gulf of St. Lawrence to the Delmarva Peninsula, with highest abundances on the Scotian Shelf and in the Gulf of Maine (Cargnelli et al. 1999). Spawning takes place from September to April on the Scotian Shelf and November to February in the Gulf of Maine (Cargnelli et al. 1999). Optimal egg development occurs between 2.2 and 8.9°C (Cargnelli et al. 1999). Eggs are pelagic and free-floating, usually found in water 50-250 m in depth. Larvae are normally occurring from the shore out to the 200 m depth contour, but have been reported in waters as deep as 1550 m (Hardy 1978). Eggs and larvae undergo significant movement away from the spawning area as they drift with currents. For example, larvae from the northern Gulf of Maine are thought to originate from spawning grounds in the southern Gulf of Maine (Hardy 1978). Larvae are pelagic, consuming generally the most abundant copepods in their preferred size range, and metamorphose into juveniles at 3-4 months of age (Cargnelli et al. 1999). Juveniles have been reported over a wide variety of substrates, including sand, mud, or rocky bottom and vegetation (Ojeda and Dearborn 1990). Inshore subtidal and intertidal zones are utilized by age 0-1 juveniles and serve as important nursery areas. Age 2+ juveniles move offshore, inhabiting depths of 130-150 m (Hardy 1978). The primary prey of juvenile pollock are crustaceans. Euphausiids, in particular *Meganyctiphanes norvegica*, are the most important crustacean prey of juveniles (Tyler 1972). Fish and mollusks make up a smaller proportion of the juvenile

diet; however, in some cases fish may play a more important role in the diet (Collette and Klein-MacPhee 2002). Pollock reach maturity at 3-7 years of age, longer for females, and reach a maximum age of 24 years (Northeast Fisheries Science Center (NEFSC) 2013). Adults inhabit a wide range of depths, 35-365 m, but most occur within the 137 m depth contour and depths of 100-125 m are preferred (Scott 1982). They prefer temperatures 0-14°C , although they tend to avoid temperatures $>$ 11°C and $<$ 3°C (Scott 1982). For most of the year, adult pollock undergo minor inshore-offshore movements linked to temperature. Pollock migration rates vary with stock areas, but there is considerable movement across the Scotian Shelf and Gulf of Maine (Mayo, McGlade, and Clark 1989). The diet of adults consists of, in order of decreasing importance, euphausiids, fish and mollusks. *M. norvegica* is the single most important prey item and Atlantic herring is the most important fish species (Steele 1963). The primary predator of adult pollock are monkfish, although spiny dogfish and other pollock also prey on adults (Cargnelli et al. 1999). Atlantic pollock are managed by the New England Fishery Management Council under the Northeast Multispecies Fishery Management Plan. The stock is not currently overfished and overfishing is not occurring (Northeast Fisheries Science Center (NEFSC) 2024).

Supplemental Scores

Exposure Timeseries

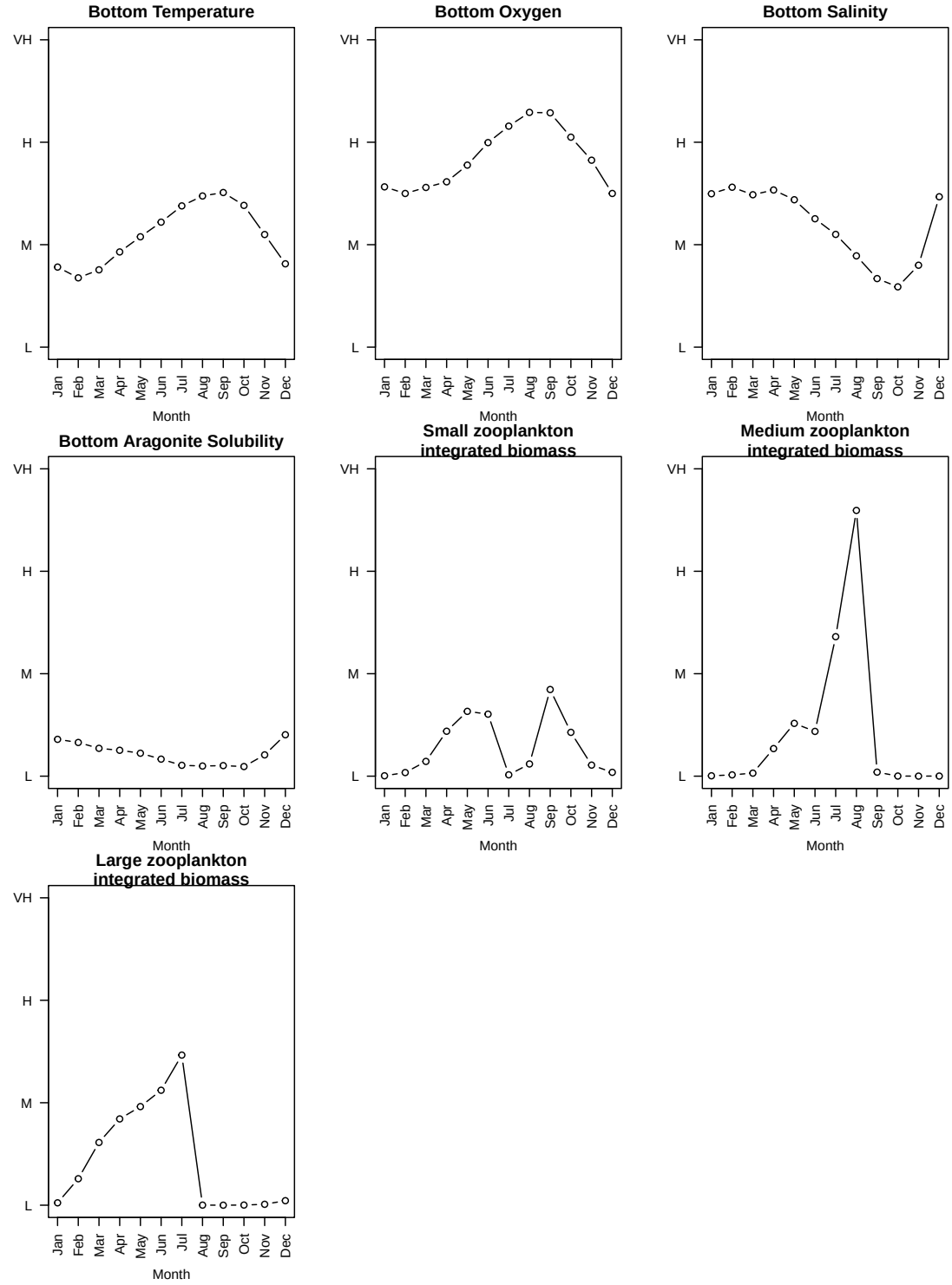


Figure 7: Monthly Timeseries of each Exposure Factor for Pollock. Exposure was calculated between 2014-2023 and 2025-2035

Exposures ranged from Low to High throughout the year, peaking in August (Figure 8). Bottom temperature, oxygen, and salinity exposure were relatively consistent throughout the year; changes were primarily due to spikes in exposure due to zooplankton biomass (Figure 7).

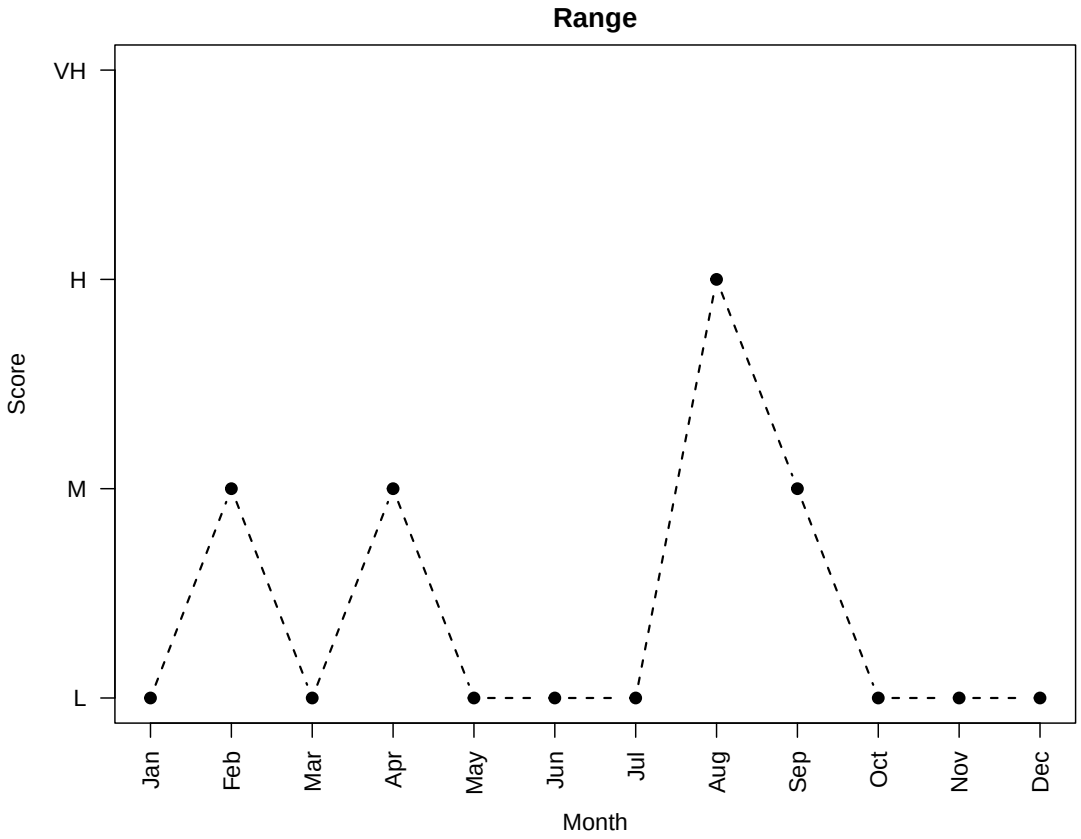


Figure 8: Timeseries of Total Exposure for Pollock

Exposure due to Significant Factors

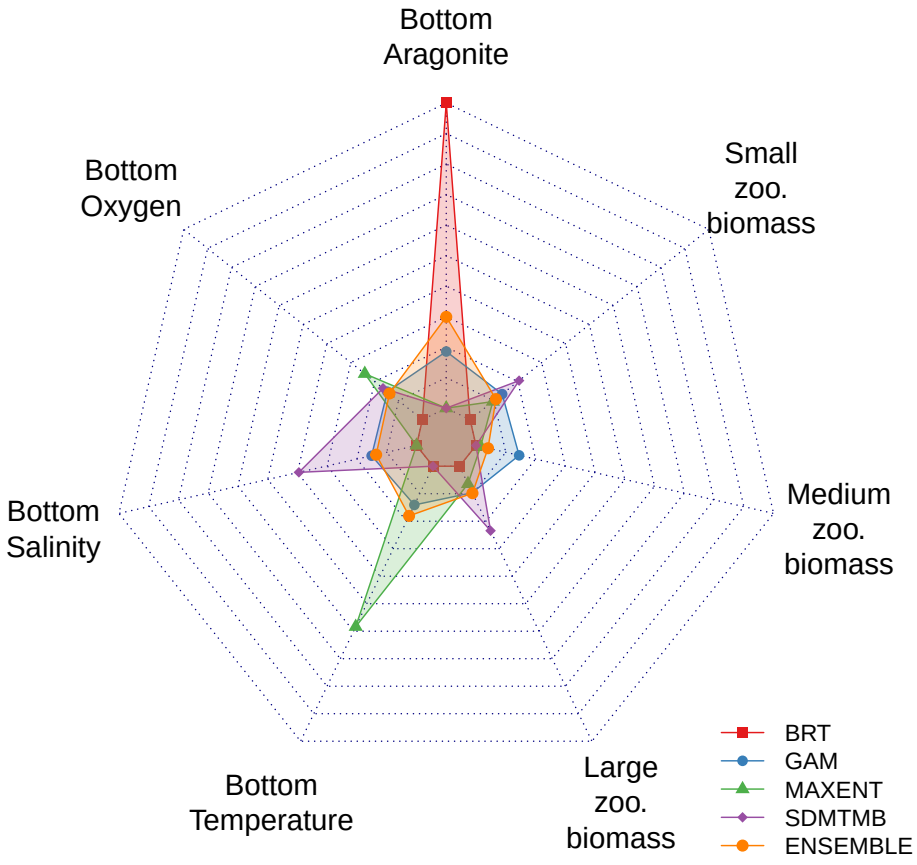


Figure 9: Dynamic Factors driving Pollock distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, salinity, and aragonite solubility, as well as small zooplankton biomass were important to pollock distributions (Figure 9). Temperature plays an important role in distribution and reproductive success (Cargnelli et al. 1999) (Collette and Klein-MacPhee 2002) (Scott 1982). Larvae feed primarily on zooplankton, and adults feed on small fish and krill (Cargnelli et al. 1999) (Steele 1963). When only these factors were used to calculate exposure, the scores remained the same across space (Figure 10; Figure 11).

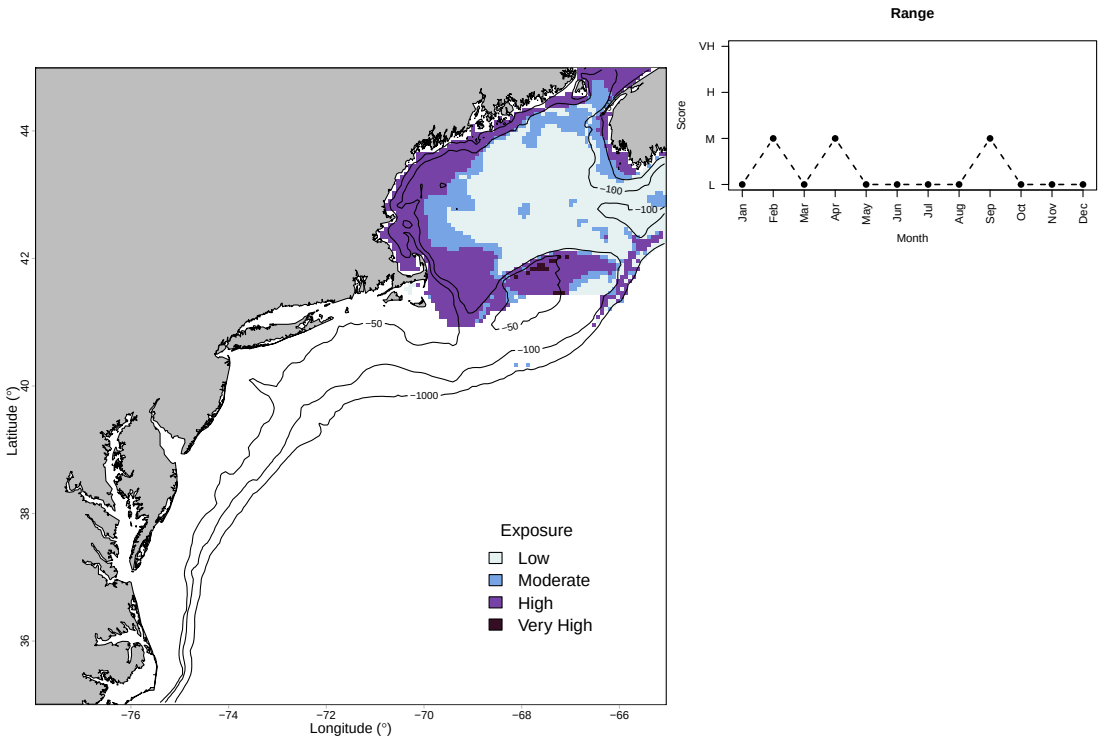


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Pollock distributions

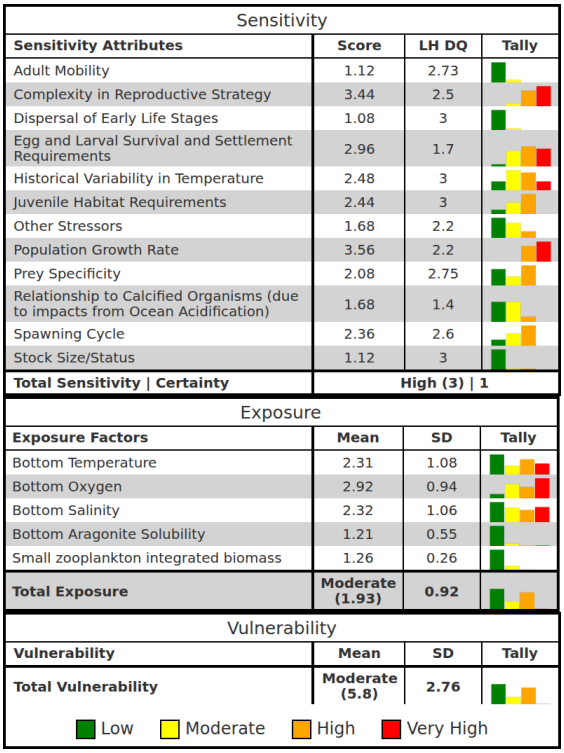


Figure 11: As in Figure 1 for Pollock, but with only significant Exposure factors and the resulting Vulnerability Scores.

References

Cargnelli, L. M., S. J. Griesbach, D. B. Packer, P. L. Berrien, D. L. Johnson, and W. W. Morse. 1999. "Essential Fish Habitat Source Document: Pollock, Pollachius Virens, Life History and Habitat Characteristics." Tech Memo 131. NOAA.

Collette, B. B., and G. Klein-MacPhee. 2002. *Bigelow and Schroeder's Fishes of the Gulf of Maine*. Washington, DC: Smithsonian Institution Press.

Hardy, J. D., Jr. 1978. "Development of Fishes of the Mid-Atlantic Bight: An Atlas of Egg, Larval and Juvenile Stages. Vol. 2 Anguillidae Through Syngnathidae." Report FWS/OBS-78/12. U.S. Fish; Wildlife Service, Biological Services Program.

Hare, Jonathan A, Wendy E Morrison, Mark W Nelson, Megan M Stachura, Eric J Teeters, Roger B Griffis, Michael A Alexander, et al. 2016. "A Vulnerability Assessment of Fish and Invertebrates to Climate Change on the Northeast US Continental Shelf." *PloS One* 11 (2): e0146756.

Hawkins, Anthony D, and Marta Picciulin. 2019. "The Importance of Underwater Sounds to Gadoid Fishes." *The Journal of the Acoustical Society of America* 146 (5): 3536–51.

Looby, Audrey, Kieran Cox, Santiago Bravo, Rodney Rountree, Francis Juanes, Laura K Reynolds, and Charles W Martin. 2022. "A Quantitative Inventory of Global Soniferous Fish Diversity." *Reviews in Fish Biology and Fisheries* 32 (2): 581–95.

Mayo, R. K., J. M. McGlade, and S. H. Clark. 1989. "Patterns of Exploitation and Biological Status of Pollock (Pollachius Virens L.) in the Scotian Shelf, Georges Bank, and Gulf of Maine Area." *Journal of Northwest Atlantic Fishery Science* 9: 13–36.

Murawski, S. A. 1993. "Climate Change and Marine Fish Distributions: Forecasting from Historical Analogy." *Transactions of the American Fisheries Society* 122 (5): 647–58.

Northeast Fisheries Science Center (NEFSC). 2013. "55th Northeast Regional Stock Assessment Workshop (55th SAW) Assessment Report." Reference Document 2013-11. Northeast Fisheries Science Center.

———. 2024. "Pollock 2024 Update Assessment Report." Northeast Fisheries Science Center. https://apps-st.fisheries.noaa.gov/sis/docServlet?fileAction=download&fileId=9883.

Nye, Janet A., Jason S. Link, Jonathan A. Hare, and William J. Overholtz. 2009. "Changing Spatial Distribution of Fish Stocks in Relation to Climate and Population Size on the Northeast United States Continental Shelf." *Marine Ecology Progress Series* 393: 111–29.

Ojeda, F. P., and J. H. Dearborn. 1990. "Diversity, Abundance, and Spatial Distribution of Fishes and Crustaceans in the Rocky Subtidal Zone of the Gulf of Maine." *Fishery Bulletin (U.S.)* 88: 403–10.

Scott, J. S. 1982. "Depth, Temperature and Salinity Preferences of Common Fishes of the Scotian Shelf." *Journal of Northwest Atlantic Fishery Science* 3: 29–39.

Steele, D. H. 1963. "Pollock (Pollachius Virens (L.)) in the Bay of Fundy." *Journal of the Fisheries Research Board of Canada* 20: 1267–1314.

Tyler, A. V. 1972. "Food Resource Division Among Northern Marine Demersal Fishes." *Journal of the Fisheries Research Board of Canada* 29: 997–1003.